BIG DATA AND THE BUSINESS CASE

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AGENDA

- Realizing Value
- The Case for Big Data
- The Rise of Big Data Options
- Beyond Hadoop
- With Choice Come Decisions

REALIZING VALUE

Revenue Growth

- Personalization, recommendation engines, targeted marketing, and dynamic pricing increase customer conversion.

Cost Efficiency

- Predictive maintenance, process optimization, and automation reduce operational expenditure.

Risk Reduction

- Fraud detection, anomaly monitoring, and early warning systems help avoid financial and reputational damage.

Customer Experience

- eReal-time engagement analytics improve service quality and retention rates.

Innovation

- Insight discovery enables new products, new business models, and data monetization opportunities.

THE CASE FOR BIG DATA

A strong case emerges when the limitations of traditional systems hinder business objectives. Big Data technologies become necessary when

Data volumes exceed database capacity
Logs, clickstreams, sensor readings, multimedia, and transactional data grow faster than conventional storage.

Data arrives at high velocity
Real-time streams from IoT devices, mobile apps, and digital platforms require streaming analytics and fast ingestion.

Data variety increases
Text, images, video, geospatial signals, semi-structured files, and unstructured logs need flexible processing frameworks.

Complex analytics are required

Machine learning workloads, graph processing, natural language processing, and recommendation systems demand distributed compute.

Business requires rapid experimentation
Big data environments support parallel model training, A/B experimentation, and iterative prototyping.

BIG DATA OPTIONS

1. **Cloud computing**: On-demand storage and compute reduce upfront capital expenditure and enable elastic scaling.
2. **Distributed processing engines**: Spark, Flink, and cloud-native services accelerate batch, streaming, and ML workloads.
3. **NoSQL databases**: Document stores, key-value stores, and columnar databases handle schema flexibility and massive read/write loads.
4. **Streaming platforms**: Kafka, Pulsar, and Kinesis process high-velocity data in real time.
5. **Managed services**: BigQuery, Snowflake, Redshift, and Databricks simplify analytics operations and reduce cluster management overhead.

BEYOND HADOOP

While Hadoop introduced scalable distributed storage and parallel processing, it is no longer the central pillar of Big Data architectures.

Speed limitations

- MapReduce's disk-heavy workflows introduce latency unsuitable for real-time analytics.

Complexity

- Cluster maintenance, tuning, and job orchestration require specialized administrators.

Emergence of in-memory engines

- Spark, Dask, and Ray support faster analytics and machine learning at scale.

Cloud-native ecosystems

- Serverless and managed data platforms provide easier scaling, reduced operational burden, and stronger integration with ML services.

WITH CHOICE COME DECISIONS

Storage Strategy
- Data lakes vs. data warehouses
- Cloud vs. on-premise
- Object storage vs. distributed file systems

Compute and Processing Engines
- Batch vs. streaming frameworks
- In-memory vs. disk-based execution
- Serverless vs. cluster-based deployment

Data Governance and Security
- Access controls, compliance (GDPR, HIPAA), data lineage
- Encryption, privacy-preserving analytics

Analytics and ML
- Toolchain selection (Spark ML, TensorFlow, PyTorch, cloud ML services)
- Model deployment and ML Ops considerations

Cost Management
- Elastic scaling vs. fixed infrastructure
- Pay-as-you-go vs. reserved resources